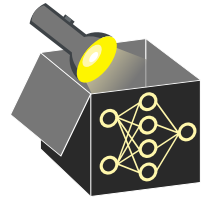
Interpretable Machine Learning



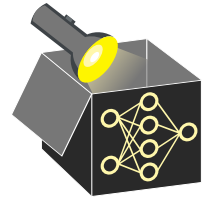
Shapley Values

Learning goals

- Learn cooperative games and value functions
- Define the marginal contribution of a player
- Study Shapley value as a fair payout solution
- Compare order and set definitions

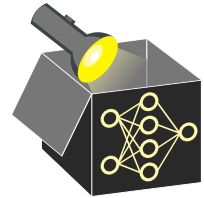
COOPERATIVE GAMES IN GAME THEORY

▸ Shapley (1951)



- **Game theory:** Studies strategic interactions among "players" (who act to maximize their utility), where outcomes depend on collective behavior
- **Cooperative games:** Any subset $S \subseteq P = \{1, \ldots, p\}$ can form a coalition to cooperate in a game, each achieving a payout $v(S)$

COOPERATIVE GAMES IN GAME THEORY ▸ Shapley (1951)

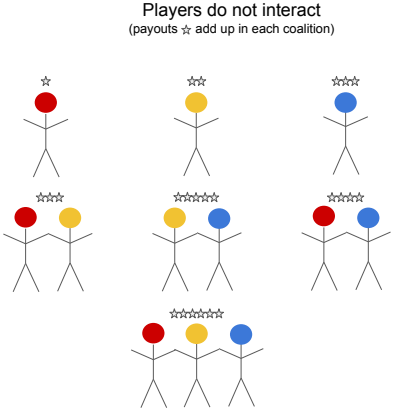


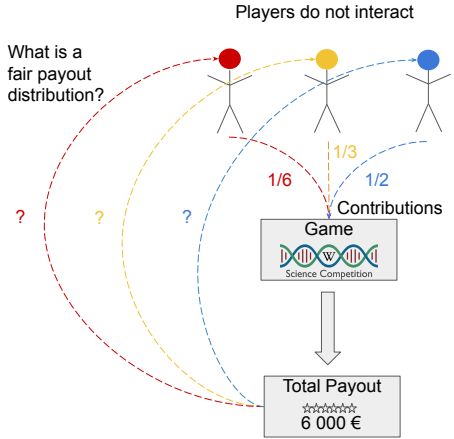
- **Game theory:** Studies strategic interactions among "players" (who act to maximize their utility), where outcomes depend on collective behavior
- **Cooperative games:** Any subset $S \subseteq P = \{1, \ldots, p\}$ can form a coalition to cooperate in a game, each achieving a payout $v(S)$
- **Value function:** $v : 2^P \to \mathbb{R}$ assigns each coalition S a payout $v(S)$
  - Convention: $v(\emptyset) = 0 \rightsquigarrow$ Empty coalitions generate no gain
  - $v(P)$: Total achievable payout when all players cooperate
    $\rightsquigarrow$ Forms the game's budget to be fairly distributed
- **Marginal contribution:** Measure how much value player j adds to coalition S by

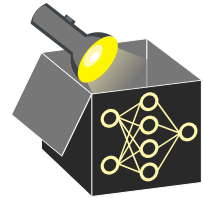
$$\Delta(j, S) := v(S \cup \{j\}) - v(S) \qquad (\text{for all } j \in P \; S \subseteq P \setminus \{j\})$$

- **Challenge:** Players vary in their contributions & how they influence each other
- **Goal:** Fairly distribute $v(P)$ among players by accounting for player interactions
  $\rightsquigarrow$ Assign each player $j \in P$ a fair share ϕ_j (**Shapley value**)

COOPERATIVE GAMES - NO INTERACTIONS

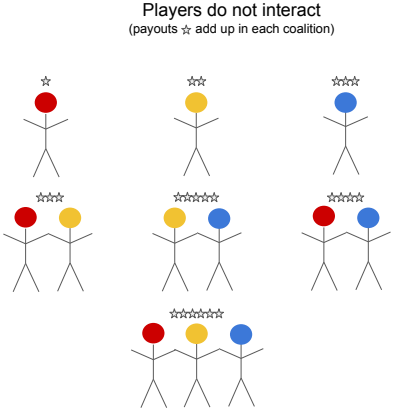


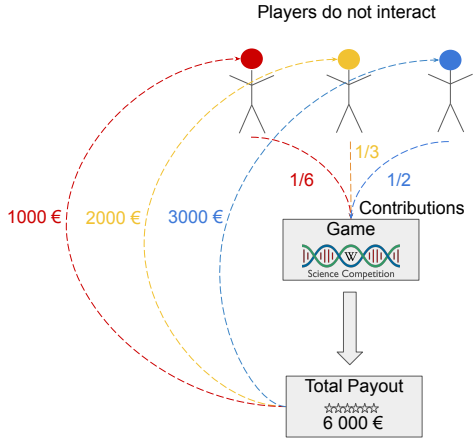


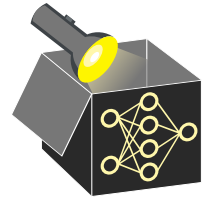


Question: What is a fair payout if contributions are $\frac{1}{2}$ (blue), $\frac{1}{3}$ (yellow), $\frac{1}{6}$ (red)?

COOPERATIVE GAMES - NO INTERACTIONS

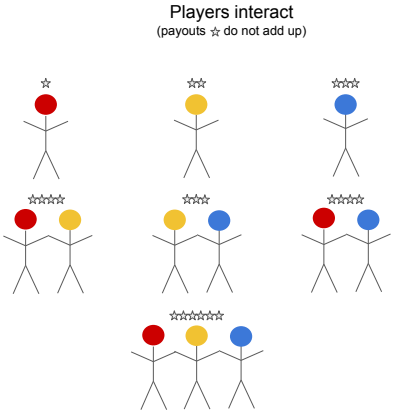


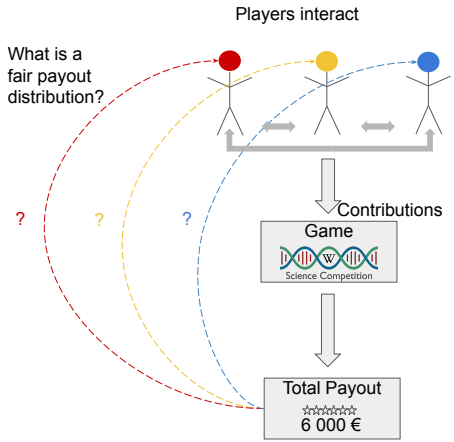


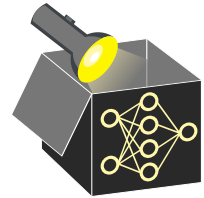


$\Rightarrow$ Fair payouts are trivial without interactions

COOPERATIVE GAMES - INTERACTIONS

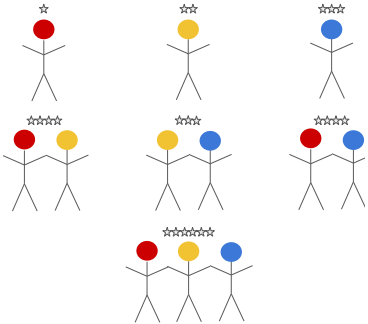






⇒ Unclear how to fairly distribute payouts when players interact

COOPERATIVE GAMES - INTERACTIONS



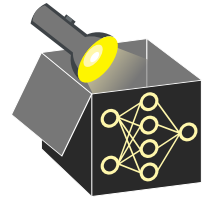
Ordering 1: 🟡 → 🔵 → 🔴

- 🟡 joins alone: 2 ☆
- 🔵 joins: total = 3 ☆, marginal = +1
- 🔴 joins: total = 6 ☆, marginal = +3

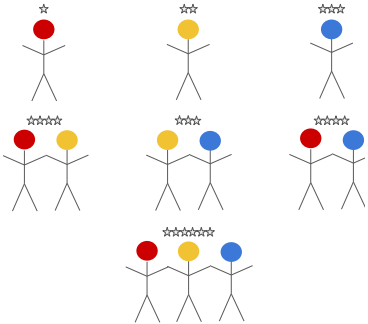
But what if 🔴 joins before 🔵?

Ordering 2: 🟡 → 🔴 → 🔵

- 🟡 joins alone: 2 ☆
- 🔴 joins: total = 4 ☆, marginal = +2
- 🔵 joins: total = 6 ☆, marginal = +2



COOPERATIVE GAMES - INTERACTIONS



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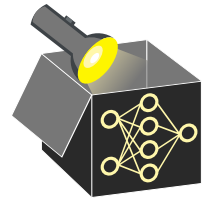
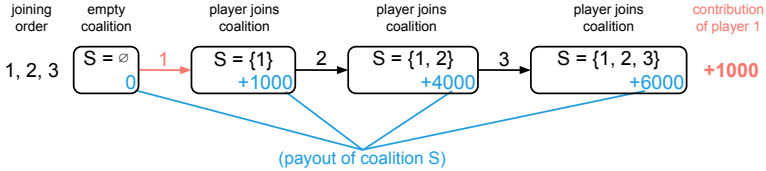


- **Overlapping skills:** Different join orders lead to different marginal contributions
- **Fairness:** Shapley values consider every possible way the team could have formed and average each player's contribution across all possible join orders
  - ⇝ Avoids redundancy (e.g., 🔵's contribution overlaps with 🔴's)
  - ⇝ Accounts for order sensitivity (e.g., 🔴 brings more value if added last)
  - ⇝ Ensures no player is advantaged or penalized by order of joining

SHAPLEY VALUES - ILLUSTRATION

- Generate all possible joining orders of players (all permutations of full set P)
- For each order: track player j-th marginal contribution when j joins a coalition



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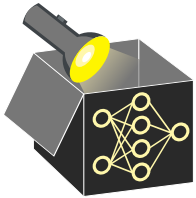
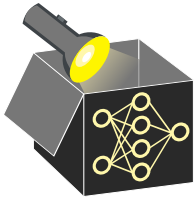


- Generate all possible joining orders of players (all permutations of full set P)
- For each order: track player j-th marginal contribution when j joins a coalition
- Shapley value of j: Average this marginal contribution over all joining orders
- **Example:** Compute payout difference after player 1 enters coalition $\rightsquigarrow$ average

joining order	empty coalition		player joins coalition		player joins coalition		player joins coalition	contribution of player 1
1, 2, 3	S = ∅ 0	1 →	S = {1} +1000	2 →	S = {1, 2} +4000	3 →	S = {1, 2, 3} +6000	**+1000**
1, 3, 2	S = ∅ 0	1 →	S = {1} +1000	3 →	S = {1, 3} +4000	2 →	S = {1, 2, 3} +6000	**+1000**
2, 1, 3	S = ∅ 0	2 →	S = {2} +2000	1 →	S = {1, 2} +4000	3 →	S = {1, 2, 3} +6000	**+2000**
2, 3, 1	S = ∅ 0	2 →	S = {2} +2000	3 →	S = {2, 3} +3000	1 →	S = {1, 2, 3} +6000	**+3000**
3, 1, 2	S = ∅ 0	3 →	S = {3} +3000	1 →	S = {1, 3} +4000	2 →	S = {1, 2, 3} +6000	**+1000**
3, 2, 1	S = ∅ 0	3 →	S = {3} +3000	2 →	S = {2, 3} +3000	1 →	S = {1, 2, 3} +6000	**+3000**

Shapley value of player 1 : **+1833.33**

SHAPLEY VALUES - ILLUSTRATION



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- For each order: track player j-th marginal contribution when j joins a coalition
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- **Example:** Compute payout difference after player 2 enters coalition $\rightsquigarrow$ average

joining order	empty coalition		player joins coalition		player joins coalition		player joins coalition	contribution of player 2
1, 2, 3	S = ∅ 0	1	S = {1} +1000	2	S = {1, 2} +4000	3	S = {1, 2, 3} +6000	+3000
1, 3, 2	S = ∅ 0	1	S = {1} +1000	3	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+2000
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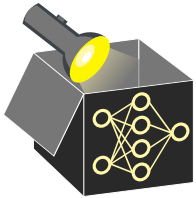
3, 2, 1	S = ∅ 0	3	S = {3} +3000	2	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+0

Shapley value of player 2 : **+1833.33**

SHAPLEY VALUES - ILLUSTRATION



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- **Example:** Compute payout difference after player 3 enters coalition $\rightsquigarrow$ average

joining order	empty coalition		player joins coalition		player joins coalition		player joins coalition	contribution of player 3
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1, 3, 2	S = ∅ 0	1	S = {1} +1000	3	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+3000
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3, 2, 1	S = ∅ 0	3	S = {3} +3000	2	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+3000

Shapley value of player 3 : +2333.33

SHAPLEY VALUES - ILLUSTRATION

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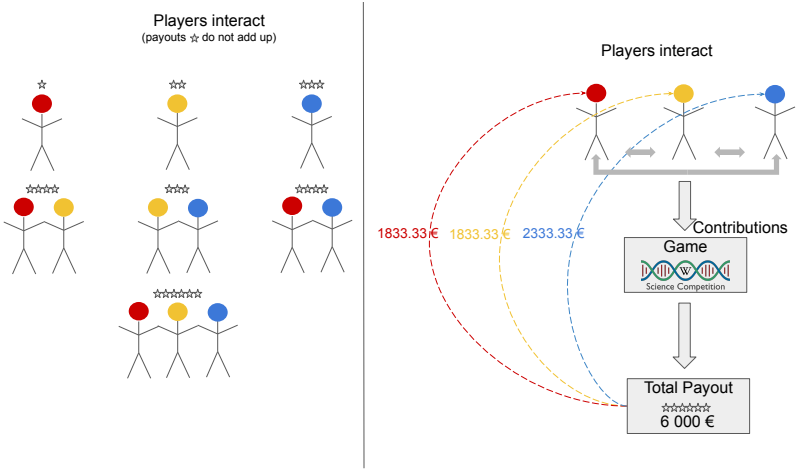
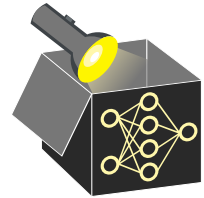
- Shapley value of j: Average this marginal contribution over all joining orders



SHAPLEY VALUE - ORDER DEFINITION

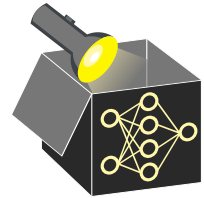


The Shapley value order definition averages the marginal contribution of a player across all possible player orderings:

$$\phi_j = \frac{1}{|P|!} \sum_{\tau \in \Pi} (v(S_j^\tau \cup \{j\}) - v(S_j^\tau))$$

- Π: Set of all permutations (joining orders) of the players – there are $|P|!$ in total

SHAPLEY VALUE - ORDER DEFINITION



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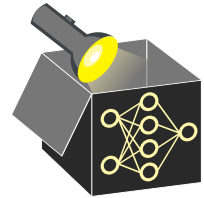
- S_j^τ: Set of players before j joins, for each ordering $\tau = (\tau^{(1)}, \ldots, \tau^{(p)})$
  For $P = \{1, 2, 3\}$**:** $\Pi = \{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)\}$
  $\rightsquigarrow$ For joining order $\tau = (2, 1, 3)$ and player of interest $j = 3 \Rightarrow S_j^\tau = \{2, 1\}$
  $\rightsquigarrow$ For joining order $\tau = (3, 1, 2)$ and player of interest $j = 1 \Rightarrow S_j^\tau = \{3\}$

SHAPLEY VALUE - ORDER DEFINITION



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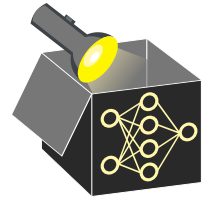
  $\rightsquigarrow$ For joining order $\tau = (3, 1, 2)$ and player of interest $j = 1 \Rightarrow S_j^\tau = \{3\}$
- Order definition allows to approximate Shapley values by sampling permutations
  $\rightsquigarrow$ Sample a fixed number $M \ll |P|!$ of random permutations and average:

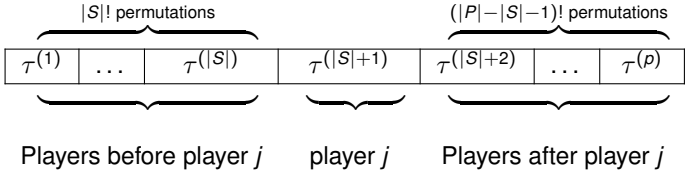
$$\phi_j \approx \frac{1}{M} \sum_{\tau \in \Pi_M} \left(v(S_j^\tau \cup \{j\}) - v(S_j^\tau)\right)$$

where $\Pi_M \subset \Pi$ is the random sample of M player orderings

FROM ORDER DEFINITION TO SET DEFINITION



- **Note:** The same subset S_j^τ can occur in multiple permutations (joining orders)
  $\rightsquigarrow$ Its marginal contribution is included multiple times in the sum in ϕ_j
- **Example (for set of players $P = \{1, 2, 3\}$, player of interest $j = 3$):**
  $\Pi = \{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)\}$
  $\rightsquigarrow$ In both $(1, 2, 3)$ and $(2, 1, 3)$, player 3 joins after coalition $S_j^\tau = \{1, 2\}$
  $\Rightarrow$ Marginal contribution $v(\{1, 2, 3\}) - v(\{1, 2\})$ occurs twice in ϕ_j
- **Reason:** Each subset S appears in $|S|!(|P| - |S| - 1)!$ orderings before j joins
  $\Rightarrow$ There are $|S|!$ possible orders of players within coalition S
  $\Rightarrow$ There are $(|P| - |S| - 1)!$ possible orders of players without S and j



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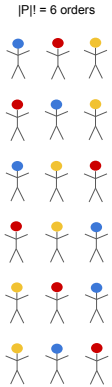
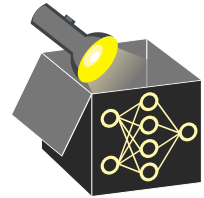
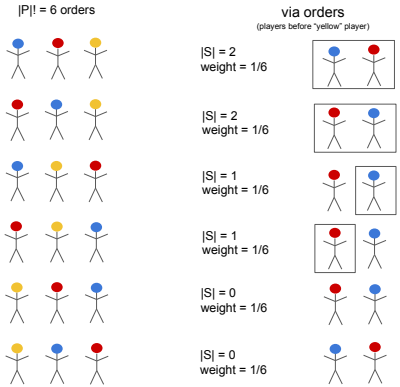
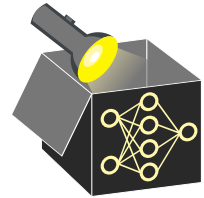




- **Order view:** Each of the $|P|!$ permutations contributes one term with weight $\frac{1}{|P|!}$
- Subsets $S \subseteq P \setminus \{j\}$ can appear before j in multiple permutations
- **Set view:** Group by unique subsets S, not permutations
- Each S occurs in $|S|!(|P| - |S| - 1)!$ orderings $\rightsquigarrow$ Weight: $\frac{|S|!(|P|-|S|-1)!}{|P|!}$

FROM ORDER DEFINITION TO SET DEFINITION





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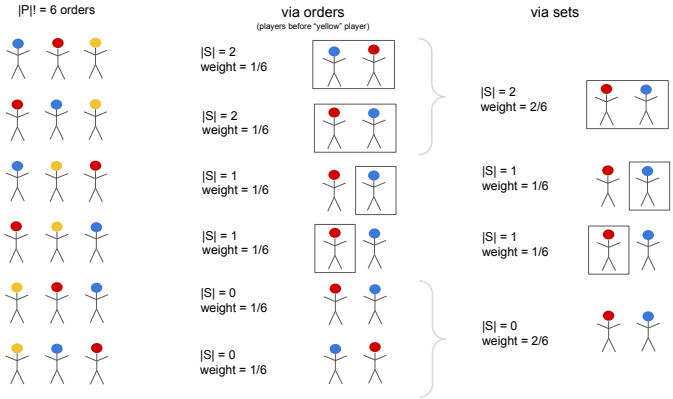
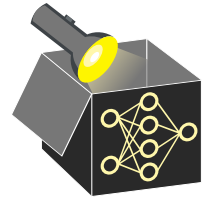
FROM ORDER DEFINITION TO SET DEFINITION





- **Order view:** Each of the $|P|!$ permutations contributes one term with weight $\frac{1}{|P|!}$
- Subsets $S \subseteq P \setminus \{j\}$ can appear before j in multiple permutations
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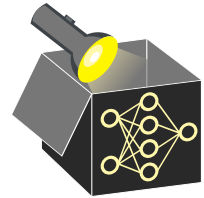
SHAPLEY VALUE - SET DEFINITION



Shapley value via **set definition** (weighting via multinomial coefficient):

$$\phi_j = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|!(|P| - |S| - 1)!}{|P|!} (v(S \cup \{j\}) - v(S))$$

The coefficient gives the probability that, when randomly arranging all $|P|$ players, the exact set S appears before player j, and the remaining players appear afterward.

$$\overbrace{\begin{array}{|c|c|c|}\hline \tau^{(1)} & \dots & \tau^{(|S|)} \\ \hline \end{array}}^{|S|!\text{ permutations}} \overbrace{\begin{array}{|c|}\hline \tau^{(|S|+1)} \\ \hline \end{array}}^{\text{player } j} \overbrace{\begin{array}{|c|c|c|}\hline \tau^{(|S|+2)} & \dots & \tau^{(|P|)} \\ \hline \end{array}}^{(|P|-|S|-1)!\text{ permutations}}$$

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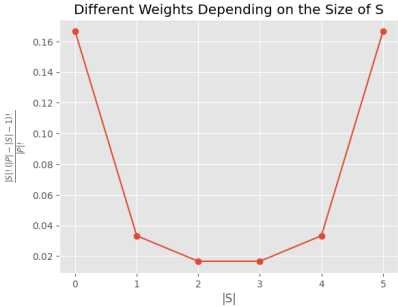
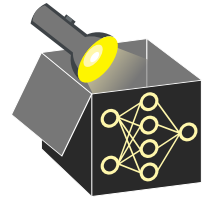


- $|S| = 0$: player j joins first
  $\Rightarrow$ many permutations $\Rightarrow$ high weight
- $|S| = |P| - 1$: player j joins last
  $\Rightarrow$ many permutations $\Rightarrow$ high weight
- Middle-sized $|S|$: fewer exact matches
  $\Rightarrow$ lower weight
- Result: U-shaped weight distribution

AXIOMS OF FAIR PAYOUTS



Why is this a fair payout solution?
One possibility to define fair payouts are the following axioms for a given value function v:

- **Efficiency**: Player contributions add up to the total payout of the game: $\sum_{j=1}^{p} \phi_j = v(P)$

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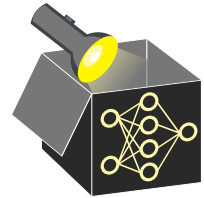

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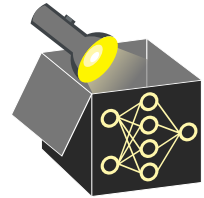

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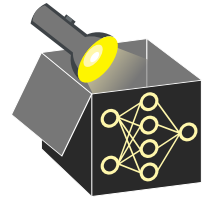

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- **Additivity**: For games with value functions v_1, v_2, their combined value function $v(S) = v_1(S) + v_2(S)$ implies that the payout $\phi_{j,v}$ is $\phi_{j,v_1} + \phi_{j,v_2}$